

One-Pager Summary: NBA Player Position Classification Using Machine Learning

Domain Background

In professional basketball, a player's position, such as Point Guard (PG), Shooting Guard (SG), Small Forward (SF), Power Forward (PF) or Center (C), plays a crucial role in team strategy and performance. However, with evolving playstyles and hybrid roles, traditional classifications are becoming blurred. Our goal was to develop a data-driven method to classify NBA players into their primary positions based on their in-game attributes and performance statistics.

Key Findings

We used a dataset from 2025 containing over 80 performance-based attributes of NBA players, including scoring ability, athleticism, defense, rebounding and size that reflect their playing style. Our analysis focused on predicting the player's main position (label position_1 in the data) using these variables. After cleaning and preparing the data, we tested various classification models including: Multinomial Logistic Regression (MLR), Support Vector Machines (SVM) and Decision Trees.

Among all models, MLR emerged as the most effective, achieving the best performance on unseen test data without signs of overfitting. While models like decision trees and SVMs were prone to overfitting on the data we were working with, the logistic regression model demonstrated a balanced and generalizable performance. It achieved a 49% improvement over the no-information rate, successfully predicting easily identifiable positions like PG and C, while having more difficulty distinguishing SG, SF and PF.

Model Suitability

The MLR model turned out to be the most reliable model for this task. It performed consistently well on both the training data and new/unseen data, which shows that it learned the patterns in the data without simply memorizing them. A major advantage of this model is that it's very interpretable, we can clearly understand how each player attribute affects the final prediction. Unlike the other black-box models we tested, MLR allows us to trace back the reasoning behind its predictions, which makes it easier to trust, explain, and improve.

Contributions and Insights

Our findings show that machine learning can provide a data-driven method to classify NBA players, even when their roles are not clearly defined by traditional standards. The success of logistic regression suggests that simpler, well-tuned models can outperform more complex ones when the dataset is well-prepared. This approach has the potential to help scouts, analysts, and coaches better understand player roles and support roster decisions.

Conclusion

Through this project, we demonstrated the effectiveness of using machine learning (particularly MLR) to classify NBA players into their primary positions. Our model offers a strong foundation for further exploration into hybrid roles, player versatility, and evolving game strategies, bringing data-driven insights into basketball team management.